

Autocorrelation & Stationarity

Box - Pierce Test, Ljung - Box Test

Q → Are there any non-zero correlations with lags one or greater.

Step 1 → Estimate acf, $\hat{\gamma}_x(h)$

autocorrelation at lag h

Step 2 → Write the Hypothesis

$$H_0 = \gamma_x(1) = \dots = \gamma_x(h) = 0$$

↑
up to some
order L

$$H_1 = \exists \gamma ; \text{ s.t. } \gamma_x(h) \neq 0$$

$$h \in \{1, \dots, L\}$$

$\hat{\gamma}$ ⇒ estimate from data

γ ⇒ Real value (needs ∞ points to calc)

1) Box - Pierce :

$$\sqrt{n} \hat{\rho}_x(1) \approx N(0, 1) \quad \text{CLT}$$

(Central Limit Theorem)

$$n \left(\hat{\rho}_x(1) \right)^2 \approx \chi^2(1)$$

$\searrow$ $\chi^2(\quad)$ = $\left(N(0, 1) \right)^2$
No. of $N(0, 1)$

Test Stat :

$$Q_{BP} = n \sum_{j=1}^h \hat{\rho}_x(j)^2 \approx \chi^2(h)$$

2) Ljung - Box Test :

$$Q_{LB} = n(n+2) \sum_{j=1}^h \frac{\hat{\rho}_x(j^2)}{n-j}$$

=> Not like one is better than other.

=> Both gives χ^2 distribution.

We don't apply these models on to time series X_t , but instead to the residual of X_t after fitting ARMA(p, q) to it.

ARMA(p, q) = AutoRegressive Moving Average model. It's a way of modeling a time series as a combination of:

- **AR part:** today's value depends on the last p values of itself (e.g. today's price $\approx$ some fraction of yesterday's price + noise)
- **MA part:** today's value also depends on the last q noise/error terms

$\Rightarrow$ We estimate p & q , so Q_{DP} or Q_{LB} tests with $\chi^2(h-p-q)$. $h > p+q$

3) Durbin - Watson Test (Autocorrelation)

$\Rightarrow$ Tests for the presence of order 1 of correlation.

Consider the time series

$$X_t = \underbrace{\beta_0 + \beta_1 t + \beta_2 t^2 + \dots + \beta_p t^p}_{\text{Deterministic}} + \underbrace{\varepsilon_t}_{\substack{\text{Residual} \\ \text{Process}}}$$

$\Rightarrow$ Compute Least Sq. Estimate $\hat{\beta}$

$\Rightarrow$ Get residuals $\hat{\varepsilon}_t = X_t - \hat{\beta} \cdot (1, t, t^2, \dots)$
 $\uparrow$ $\uparrow$
Obs. Predicted

Assume the model

$$\hat{r}_t = \phi \hat{r}_{t-1} + u_t$$

that is an AR(1) model for the residuals

✓ No autocorrelation

Then test $H_0: \phi = 0$

$$H_1: \phi \neq 0$$

↪ Yes Autocorrelation

$$\text{Test Stat: } Q_{DW} = \frac{\sum_{t=2}^T (\hat{r}_t - \hat{r}_{t-1})^2}{\sum_{t=1}^T \hat{r}_t^2}$$

$$Q_{DW} \in [0, 4]$$

• If Q_{DW} is close to zero, then

$$\hat{r}_t \text{ is close to } \hat{r}_{t-1}$$

∴ +ve autocorrelation

• If Q_{DW} is close to 4, then

$$\hat{r}_t \text{ is close to } -\hat{r}_{t-1}$$

∴ -ve autocorrelation

- If ρ_{DW} is close to 2 then, we don't have any autocorrelation.

Note: a p -value can be computed but the distribution is a linear combination of χ^2 random variables

The setup: You have a null hypothesis H_0 — some boring "default" claim, like "there's no autocorrelation" or "this coin is fair." You collect data and compute some number from it (a test statistic).

The question p -value answers: "If H_0 were actually true, how surprising is the data I just saw?"

Concrete example — coin flips: H_0 = "the coin is fair (50/50)." You flip it 100 times, get 65 heads. The p -value is: *if the coin really were fair, what's the probability of getting 65 or more heads (or equivalently extreme) just by random chance?* If that probability is tiny (say, 0.001), it means "a fair coin basically never produces this by accident" — so you doubt H_0 and lean toward "this coin probably isn't fair." If that probability is large (say, 0.4), it means "eh, a fair coin does this pretty often" — no real evidence against H_0 .

Small p -value = your data would be weird/rare if H_0 were true $\rightarrow$ reject H_0 .

Large p -value = your data is totally normal under $H_0 \rightarrow$ no reason to reject it.

That threshold is usually 0.05 by convention — $p < 0.05 \rightarrow$ reject H_0 .

4) Breusch - Godfrey

$\Rightarrow$ Same as Durbin Watson, but it does it for $AR(p)$ models on the residue process $\hat{\eta}_t$. (Shows beyond lag 1).

$\Rightarrow$ Mathematically, we fit a linear Model (least square Regression), get the R^2 value.

Least squares regression — the idea

Say you have data points (x, y) — like x = ICICI Bank price, y = HDFC Bank price, one pair per day. You want to draw the "best fit" straight line through this scatter of points: $\hat{y} = a + bx$ (a = intercept, b = slope).

"Best fit" needs a precise definition, and least squares gives you one: for any candidate line, look at the vertical gap between each real point y , and what the line predicts, $\hat{y}$. That gap ($y - \hat{y}$) is called a **residual** — literally "leftover, unexplained by the line." Square each residual (so positive and negative gaps don't cancel out, and big misses get punished harder than small ones), then add them all up. That sum is called the **residual sum of squares**.

Least squares regression = pick the a and b that make that sum as small as possible. That's it — it's an optimization problem, minimize total squared error. This is exactly the same OLS you'll use to compute your static hedge ratio in the pairs trading project (regressing one bank's price on another's).

Now, R^2 — how good is that line?

R^2 answers: "of all the variation in y , how much did my line actually explain, versus how much is still unexplained noise?"

Three quantities, all sums of squares:

- **Total Sum of Squares (TSS)** — how spread out the actual y , values are around their own average $\bar{y}$, ignoring the line entirely. This is "total variation that exists in y ."
- **Residual Sum of Squares (RSS)** — the leftover squared errors after fitting the line (from above). This is "variation the line failed to explain."
- **Regression Sum of Squares** — the gap between those two: $TSS - RSS$. This is "variation the line successfully explained."

So $R^2 = \text{Regression SS} / \text{Total SS} = (\text{variation explained by the line}) / (\text{total variation that existed})$.

- $R^2 = 1 \rightarrow$ the line explains everything, every point sits exactly on it, zero residual error.
- $R^2 = 0 \rightarrow$ the line explains nothing, you'd have done just as well predicting $\bar{y}$ for every point.
- $R^2 = 0.85$ (typical "good fit" language) $\rightarrow$ 85% of the variation in y is explained by your line/ x , 15% is unexplained noise.

$\Rightarrow$ the BG test compares nR^2 to a $\chi^2(k)$
(k is the order from $AR(k)$ model)
In R, `bytest`, it can alternatively use
the standard F-test from Linear
Regression.

$\Rightarrow$ Basically, finding pattern in residuals.

Testing for Stationarity

Remember mean-reversion vs random walk from the OU process discussion? That's exactly this distinction:

- **Stationary** = mean-reverting, like the OU process — always gets pulled back toward some fixed value μ , no matter how far it strays.
- **Non-stationary** = random walk, like GBM — no pull-back force, wanders freely, today's level depends entirely on where it happened to drift to.

The following are "Unit Root Tests"

Think of a simple rule: "today's value = some fraction of yesterday's value, plus a bit of random noise." That fraction is the key knob controlling everything.

- If that fraction is **less than 1** (say 0.7), then every day, the process pulls itself back toward zero a little — yesterday's value gets shrunk before today's noise is added. Over time this settles into a stable, mean-reverting pattern — bounded, doesn't drift off. This is the stationary case.
- If that fraction is **exactly 1**, there's no shrinking at all — today's value = yesterday's value + noise, full stop. Nothing pulls it back, so it just wanders wherever the accumulated noise takes it, forever. This is the random walk case — non-stationary.
- If it's **greater than 1**, it actually explodes — grows without bound, blowing up exponentially (rarely seen in real financial data, but mathematically the third case).

Now the technical term — "root":

That "fraction" is usually called ϕ (phi) in the AR(1) model: $X_t = \phi X_{t-1} + e_t$. When you write this model as a polynomial equation ($1 - \phi z = 0$) and solve for z , the solution is $z = 1/\phi$. This z is called the root of the equation.

Here's the (slightly annoying) inversion to keep straight: because $z = 1/\phi$, the conditions flip —

- $\phi < 1 \rightarrow \text{root } z > 1 \rightarrow \text{stationary}$
- $\phi = 1 \rightarrow \text{root } z = 1 \rightarrow \text{unit root} \rightarrow \text{non-stationary}$ (this is literally where the name "unit root" comes from — the root equals exactly 1)
- $\phi > 1 \rightarrow \text{root } z < 1 \rightarrow \text{explosive}$

Connecting to ADF: the entire Augmented Dickey-Fuller test is just formally checking "is this root equal to 1 (unit root, non-stationary) or is it bigger than 1 (stationary)?" — H_0 = unit root ($z=1, \phi=1$), and rejecting H_0 means you've found evidence $\phi < 1$, i.e., the series actually mean-reverts.

1) Dickey-Fuller Test

Consider the AR(1) process,

$$\text{If } X_t = \phi X_{t-1} + \omega_t$$

$$\text{Then, } \Delta X_t = \underbrace{(\phi - 1)}_{\phi'} X_{t-1} + \omega_t$$

Want to test $H_0 : \phi' = 0$ $H_1 : \phi' \neq 0$
 $\Rightarrow H_0 : \phi = 1$ $H_1 : \phi \neq 1$
Non-stationary Stationary

$\Rightarrow$ It's done by estimating $\hat{\phi}'$

Testing does $\Phi'(z) = 1 - \phi z$ have a root at $z = 1$

* Augmented Dickey-Fuller Test
 $\Rightarrow$ Do the same thing for $AR(p)$ model.

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} + u_t$$

$$\nabla X_t = \phi_1' X_{t-1} + \sum_{j=1}^{p-1} \phi_{j+1}' \nabla X_{t-j} + u_t$$

Where $\phi_1' = \sum_{j=1}^p \phi_{j-1}$ and

$$\phi_j' = - \sum_{j=1}^p \phi_j \quad \text{for } j \geq 2$$

$$= \left[\sum_{j=1}^p \phi_{j-1} \right] X_{t-1} - \sum_{j=1}^{p-1} \left[\sum_{j=1}^p \phi_j \right] [X_{t-j} - X_{t-j+1}]$$

Note that $\Phi(z) = 1 - \sum_{j=1}^p \phi_j z^j$

So if $z = 1$, then $\Phi(1) = 0$

$\Rightarrow$ Then Test $H_0 : \phi_1' = 0$ $H_1 : \phi_1' \neq 0$

In python too, we can use an adf test where it removes the linear trend before performing the test.

$$X_t = \beta_0 + \beta_1 t + \sum_{i=1}^p \phi_i X_{t-i} + \omega_t$$

2) Phillips - Perron Test

- Tries to be more robust to deviations, from Linear Regression assumptions.

Heteroskedasticity, plain English: it means "the amount of noise/error isn't constant over time — sometimes it's small, sometimes it's big."

Contrast with **homoskedasticity** (the "nice" assumption regression usually wants): the error/noise has roughly the same size/spread throughout the whole series.

Concrete picture with your stock data: during calm market periods, day-to-day price swings might be small and steady. During a crash or high-volatility period (like COVID 2020), those swings suddenly get huge. That changing "noise size" over time is heteroskedasticity — literally "hetero" (different) + "skedasticity" (spread/scatter).

Why it matters here: the standard ADF test's math assumes errors behave "nicely" (roughly constant variance, uncorrelated). If your real data violates that — noise size changes over time, or errors are correlated with each other — ADF's results can become unreliable. The **Phillips-Perron test** does the same job as ADF (test for a unit root) but adjusts its math to still work correctly even when heteroskedasticity or correlated errors are present. So it's essentially "a more robust ADF" for messier, more realistic financial data — which is genuinely relevant for you, since stock volatility clustering (calm periods vs turbulent periods) is very real in bank stock data. This connects directly to GARCH from earlier, by the way — GARCH is literally a model of that changing volatility over time.